Jeffrey Wu

Senior Data Engineer

jeffrey.w1991@outlook.com • Austin, Texas, United States • +1(346)704-3234 • https://www.linkedin.com/in/wujeffreyl/

Summary

Senior Data Engineer with 13 years of experience designing and delivering scalable data platforms across healthcare, telecommunications, and financial services domains. Specialized in building modern data lakehouse solutions using Azure-native technologies, focusing on robust data pipelines, dimensional modeling, and secure integrations. Proven track record in translating complex business requirements into reliable, production-grade data systems that improve analytics performance and operational efficiency.

Skills

- **Programming Languages:** SQL, Python, PySpark, Scala
- **Cloud Platforms & Infrastructure:** Microsoft Azure, Azure Data Factory, Azure Synapse Analytics, Azure Databricks, Azure SQL Database, Azure Data Lake Storage (ADLS), Azure Functions
- **Data Storage & Query Systems:** SQL Server, Snowflake, Synapse SQL Pools, Delta Lake, MongoDB
- **Data Processing & Compute:** Apache Spark, Databricks, distributed data processing, batch and streaming pipelines
- **Orchestration & Workflow Management:** Azure Data Factory pipelines, metadata-driven orchestration, scheduling frameworks
- **Data Libraries & SDKs:** Pandas, PySpark APIs, REST APIs, OAuth 2.0 integrations, JSON parsing libraries
- **Data Engineering Practices & Architecture:** Data lakehouse architecture, dimensional modeling (star schema), ETL/ELT design, data quality frameworks, data governance, API ingestion, CI/CD pipelines, data security and access control

Work Experience

Senior Data Engineer, Charter Communications — Sep 2023 – Present

- Worked on a large-scale broadband analytics platform supporting customer usage insights and network performance monitoring by building end-to-end **Azure Data Factory** and **Azure Databricks** pipelines, enabling consistent ingestion and transformation of structured and semi-structured data into a centralized **ADLS** lakehouse.
- Addressed fragmented reporting and inconsistent data definitions by implementing standardized **ETL/ELT pipelines** using **PySpark** and **SQL**, applying reusable transformation logic and schema enforcement techniques to improve downstream data reliability and reduce reconciliation issues.
- Built scalable ingestion pipelines for external APIs using **OAuth 2.0** authentication and JSON parsing strategies, normalizing nested structures into relational models within **Azure Synapse**, ensuring compatibility with enterprise reporting tools.
- Designed and optimized dimensional models using **star schema** techniques in **Azure SQL Database**, aligning fact and dimension tables with business KPIs and reducing query latency for analytics workloads.
- Enhanced pipeline observability and fault tolerance by implementing metadata-driven orchestration and monitoring within **ADF**, enabling automated retries, alerting mechanisms, and improving pipeline success rates.
- Improved data security and governance by integrating role-based access control and data masking strategies across **Azure** services, ensuring compliance with internal data policies and protecting sensitive customer information.
- Contributed to performance tuning of distributed workloads in **Spark** by optimizing partitioning strategies and caching mechanisms, reducing processing time for large datasets by approximately 25%.
- Mentored junior engineers through code reviews and knowledge-sharing sessions focused on **data modeling**, **pipeline design**, and **Azure best practices**, helping improve code consistency and onboarding efficiency.

Senior Data Engineer, Kestra Financial — Jul 2021 – Aug 2023

- Developed a modern wealth analytics platform by transitioning legacy **SSIS** workflows and on-prem **SQL Server** systems into cloud-native pipelines using **Azure Data Factory** and **Databricks**, improving scalability and simplifying maintenance.
- Implemented a unified data model using **dimensional modeling** techniques in **Snowflake**, aligning financial reporting requirements with optimized query performance for advisor and client reporting dashboards.
- Built automated data validation and testing frameworks using **Python** and **SQL**, ensuring data consistency across ingestion and transformation layers while reducing production data defects.
- Integrated third-party financial APIs using secure authentication mechanisms and JSON transformation logic, enabling seamless ingestion of portfolio and transaction data into the centralized data warehouse.
- Designed a separate internal reporting product leveraging lightweight APIs and cached datasets, improving accessibility of operational metrics for internal stakeholders without impacting core data pipelines.
- Strengthened infrastructure reliability by implementing CI/CD pipelines and environment isolation strategies, ensuring consistent deployment across development and production environments.

Data Engineer, SafeAuto — Jan 2021 – Jul 2021

- Supported migration of insurance analytics systems from on-prem environments to **Azure**, building scalable ingestion and transformation pipelines using **Azure Data Factory** and **Azure SQL Database** to modernize reporting workflows.

- Converted legacy **SSRS** reports into **Power BI** dashboards by redesigning data models and optimizing SQL queries, improving reporting usability and reducing manual report generation time.
- Developed a tagging and classification framework using **MongoDB** to organize semi-structured insurance data, enabling more flexible querying and segmentation for underwriting analytics.
- Built a secondary product for claims anomaly detection using rule-based logic and batch processing pipelines, helping identify suspicious patterns in claims data for further review.
- Improved pipeline reliability by implementing retry logic and logging mechanisms within **ADF**, ensuring stable execution across multiple data sources and reducing operational interruptions.
- Collaborated in Agile sprints with analysts and engineers, contributing to documentation, backlog refinement, and iterative delivery of data features aligned with business requirements.

Data Engineer, Booz Allen Hamilton — Mar 2020 – Jan 2021

- Worked on a government data modernization initiative by building secure **ETL pipelines** using **Python** and **SQL**, enabling ingestion and transformation of large datasets into structured reporting systems.
- Designed data transformation workflows that standardized disparate datasets into unified schemas, improving interoperability across internal analytics systems.
- Developed a secondary reporting system for operational metrics tracking using lightweight APIs and scheduled batch jobs, enhancing visibility into system performance.
- Applied security best practices including encryption and access control in data pipelines, ensuring compliance with federal data protection standards.
- Improved data processing efficiency by optimizing SQL queries and indexing strategies, reducing query execution times for large datasets.
- Participated in Agile delivery cycles, collaborating with cross-functional teams to refine requirements and deliver incremental data features.

Data Engineer, Take2 Consulting, LLC — Sep 2019 – Mar 2020

- Contributed to enterprise data integration projects by building **SQL-based ETL pipelines** that consolidated data from multiple operational systems into centralized reporting databases.
- Developed transformation logic for structured datasets, ensuring data consistency and improving reporting accuracy across business units.
- Built an internal analytics tool for tracking project performance metrics, leveraging simple APIs and scheduled jobs for data updates.
- Optimized database performance through indexing and query tuning, improving response times for reporting queries.
- Supported deployment processes by implementing basic CI/CD workflows and version control practices for data scripts.
- Collaborated with stakeholders to gather requirements and translate them into technical data solutions aligned with business needs.

Data Engineer, Q2ebanking — Jun 2013 – Sep 2019

- Worked on fintech banking platforms by developing **SQL-based data pipelines** that processed transactional and customer data for reporting and analytics purposes.
- Designed and maintained data models supporting digital banking features, ensuring data integrity and consistency across systems.
- Built a secondary analytics module for monitoring user engagement metrics, improving visibility into customer behavior trends.
- Improved system reliability by implementing data validation checks and monitoring scripts for critical data processes.

Education

Bachelor’s of Mathematics Actuarial Science, The University of Texas at Austin — 2009 - 2013